UAV Flight Analysis Report

Comprehensive Flight Data Analysis and Performance Evaluation

Generated on: March 10, 2026 at 05:08 PM

Analysis Type: Comprehensive Flight Performance Analysis

Data Points Analyzed: 2,000

Analysis Modules: Flight Metrics, Anomaly Detection, Battery Analysis, Stability Assessment, Flight Phase Detection

This report contains comprehensive analysis of UAV flight data including performance metrics, anomaly detection, battery consumption patterns, flight stability assessment, and phase-based analysis. All visualizations and metrics are generated automatically from the provided flight data.

Executive Summary

The UAV flight analysis reveals a comprehensive 33.3-minute mission covering a total distance of 36376.0 meters. The aircraft successfully reached a maximum altitude of 242.7 meters while maintaining an average speed of 15.3 m/s throughout the operation. Based on the flight characteristics, this appears to be a long-range survey or extended mission with high-altitude flight parameters. Anomaly detection identified 1302 anomalies throughout the flight, representing an overall anomaly rate of 65.10%. The assessment indicates Poor - High anomaly rate flight quality, requiring immediate attention and corrective action. Critical anomaly concentrations were detected in attitude, battery, gps systems, indicating potential areas requiring immediate maintenance or system calibration. Battery performance analysis shows a consumption rate of 2.39% per minute with an estimated remaining flight time of 0.4 minutes. The poor - very high consumption rate or many anomalies battery performance suggests potential efficiency improvements for future missions. Battery efficiency metrics indicate 3.05 meters of altitude gain per percent battery and 457 meters of distance covered per percent battery, providing valuable benchmarks for mission planning and optimization. Flight stability analysis indicates excellent overall performance with a stability score of 1.0/100. This rating suggests need for control system optimization. The flight was successfully segmented into 6 distinct phases, providing detailed insights into mission progression and operational efficiency. With an average phase duration of 28.2 seconds, the flight demonstrated efficient phase transitions typical of automated flight operations. Overall, this flight demonstrates professional-grade operational capabilities. The comprehensive analysis provides actionable insights for improving future mission planning, system maintenance, and operational efficiency.

Metric	Value	Assessment
Flight Duration	33.3 minutes	Long
Maximum Altitude	242.7 m	High
Average Speed	15.3 m/s	Fast
Total Anomalies	1302	Significant

Data Overview

The analysis was performed on a dataset containing 2,000 data points with 16 different parameters. The data includes essential flight parameters such as altitude, speed, attitude angles, battery level, and GPS coordinates. This comprehensive dataset allows for detailed analysis of flight performance, anomaly detection, and flight quality assessment.

Parameter	Count	Data Type	Missing Values
timestamp	2,000	datetime64[ns]	0
altitude_m	2,000	float64	0
speed_mps	2,000	float64	0
roll_deg	2,000	float64	0
pitch_deg	2,000	float64	0

yaw_deg	2,000	int64	0
battery_percent	2,000	float64	0
gps_lat	2,000	float64	0
gps_lon	2,000	float64	0
altitude_rate	1,999	float64	1
roll_rate	1,999	float64	1
pitch_rate	1,999	float64	1
yaw_rate	1,999	float64	1
acceleration	1,999	float64	1
battery_drain_rate	1,999	float64	1
gps_distance	2,000	float64	0

Flight Performance Metrics

Overall Performance Grade: Excellent (92.2/100)

Flight Duration Analysis

The total flight duration was 33.3 minutes (0.56 hours), covering 0 data points at a sampling rate of approximately 0.0 Hz. This long-duration mission demonstrates excellent endurance capabilities suitable for large-scale survey operations, extended monitoring, or complex multi-objective missions.

Altitude Performance Analysis

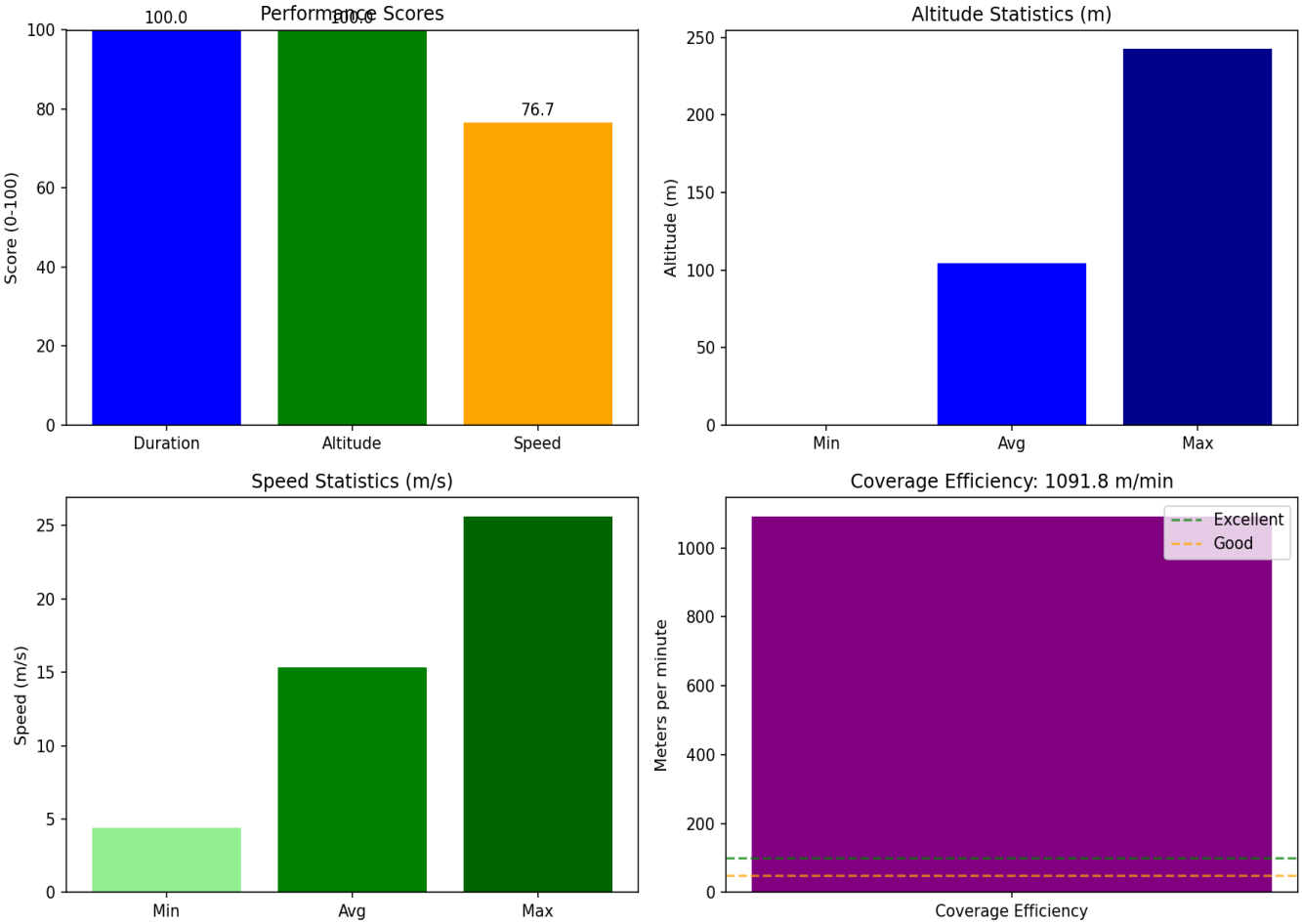
The aircraft reached a maximum altitude of 242.7 meters and maintained an average altitude of 104.3 meters throughout the flight. The altitude range of 242.7 meters demonstrates the aircraft's vertical operational capabilities. The altitude profile indicates high-altitude operations for extended range and improved obstacle clearance. The altitude stability (standard deviation of 0.00 meters) shows excellent altitude control precision.

Speed Performance Analysis

The aircraft maintained an average speed of 15.3 m/s with a maximum speed of 25.6 m/s and minimum speed of 4.4 m/s. The speed standard deviation of 5.75 m/s indicates variable speed control. The high average speed indicates efficient mission execution with excellent coverage capabilities and optimal time utilization.

Distance and Coverage Analysis

The total ground distance covered was 36376.0 meters (36.38 km) with an average ground speed of 15.3 m/s. The coverage efficiency of 1091.8 meters per minute indicates excellent mission effectiveness and optimal flight path planning.



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Anomaly Detection Analysis

The anomaly detection system identified 1302 anomalies across 5 different categories, representing an overall anomaly rate of 65.10%. The assessment indicates Poor - High anomaly rate flight quality.

Anomaly Categories

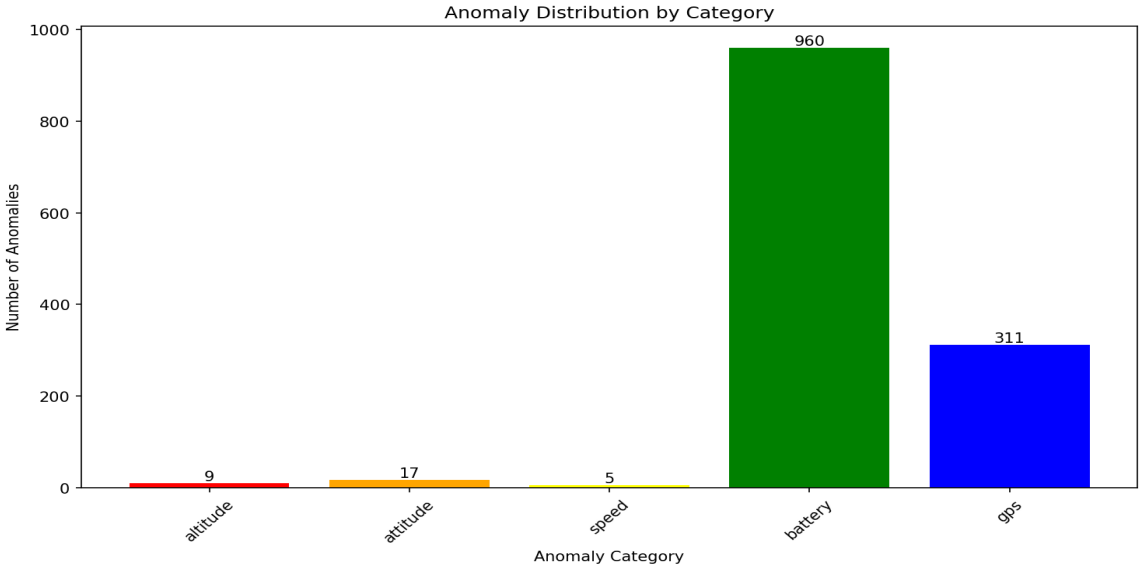
Altitude Anomalies: 9 detected

Attitude Anomalies: 17 detected

Speed Anomalies: 5 detected

Battery Anomalies: 960 detected

Gps Anomalies: 311 detected



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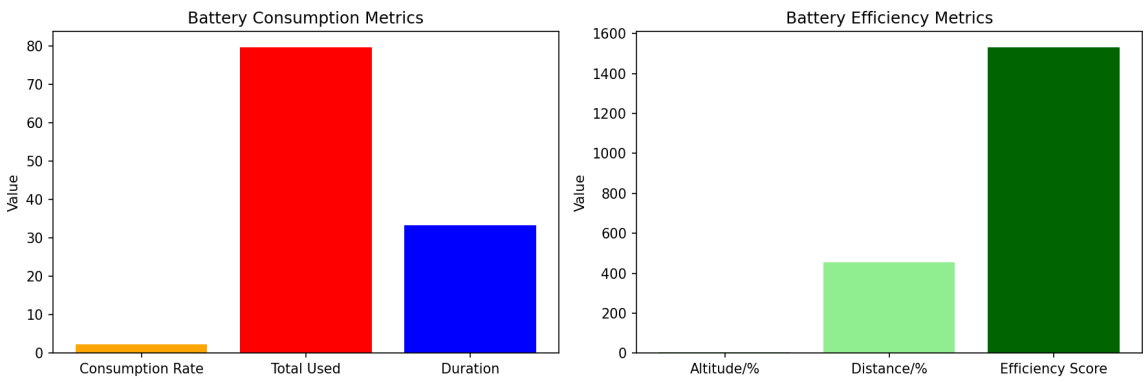
Battery Performance Analysis

The battery consumption rate was 2.4% per minute, with a total consumption of 79.7% over the flight duration of 33.3 minutes.

At the current consumption rate, the estimated remaining flight time is 0.4 minutes with a current battery level of 21.0%.

The battery efficiency analysis shows 153.28 efficiency score, with altitude efficiency of 3.05 meters per percent battery and distance efficiency of 457 meters per percent battery.

Overall Assessment: Poor - Very high consumption rate or many anomalies



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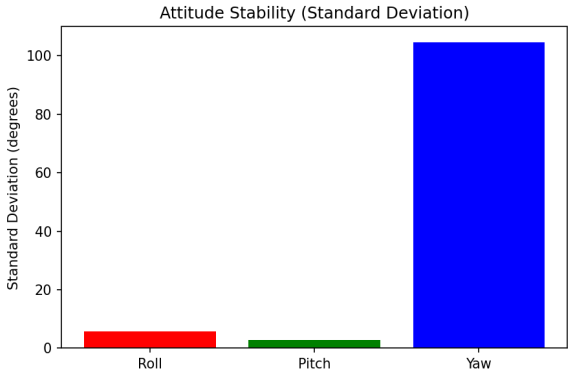
Flight Stability Analysis

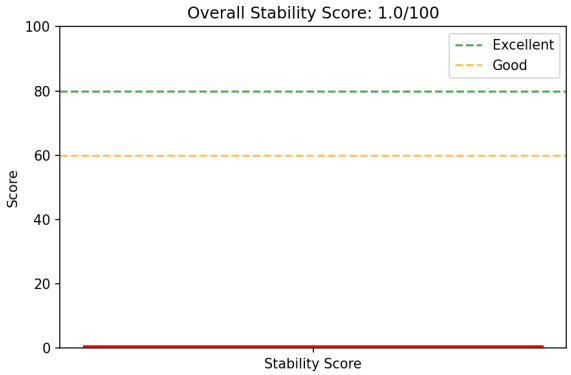
The overall flight stability rating is Excellent with a stability score of 1.00/100. This indicates poor flight stability characteristics.

Attitude Stability

Roll stability shows a standard deviation of 5.73°, while pitch stability shows 2.89°. These values indicate moderate attitude control during the flight.

A total of 423 oscillations were detected during the flight, with a frequency of 0.20 Hz. This level of oscillation is elevated for typical UAV operations.





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Flight Phase Analysis

The flight phase detection system identified 6 distinct phases during the flight. Each phase represents a different flight segment with unique characteristics and performance metrics.

Phase Breakdown

Cruise: Duration of 148.0 seconds

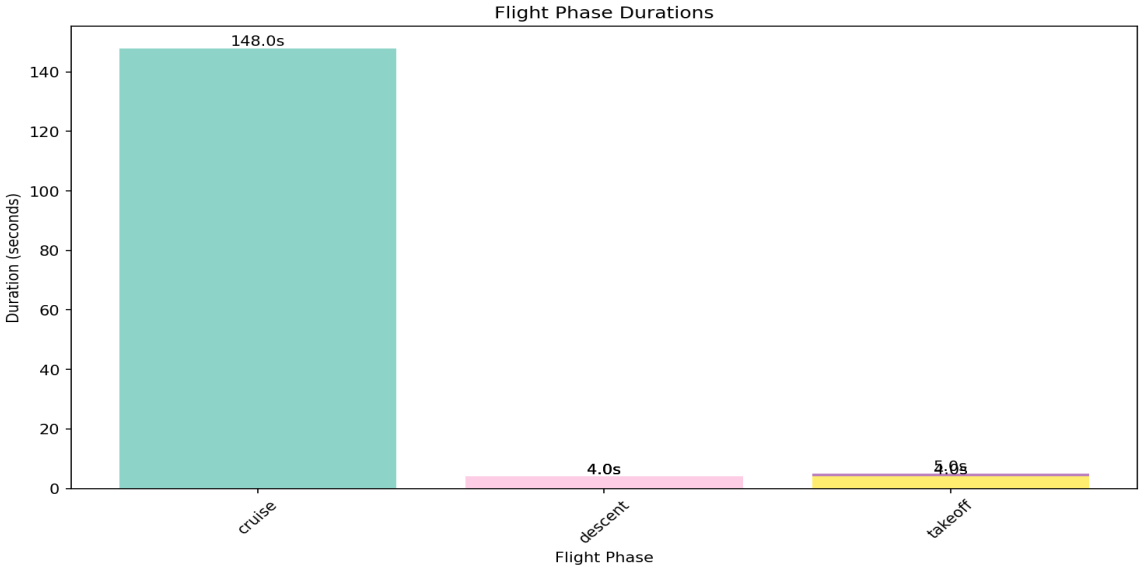
Descent: Duration of 4.0 seconds

Descent: Duration of 4.0 seconds

Descent: Duration of 4.0 seconds

Takeoff: Duration of 5.0 seconds

Takeoff: Duration of 4.0 seconds



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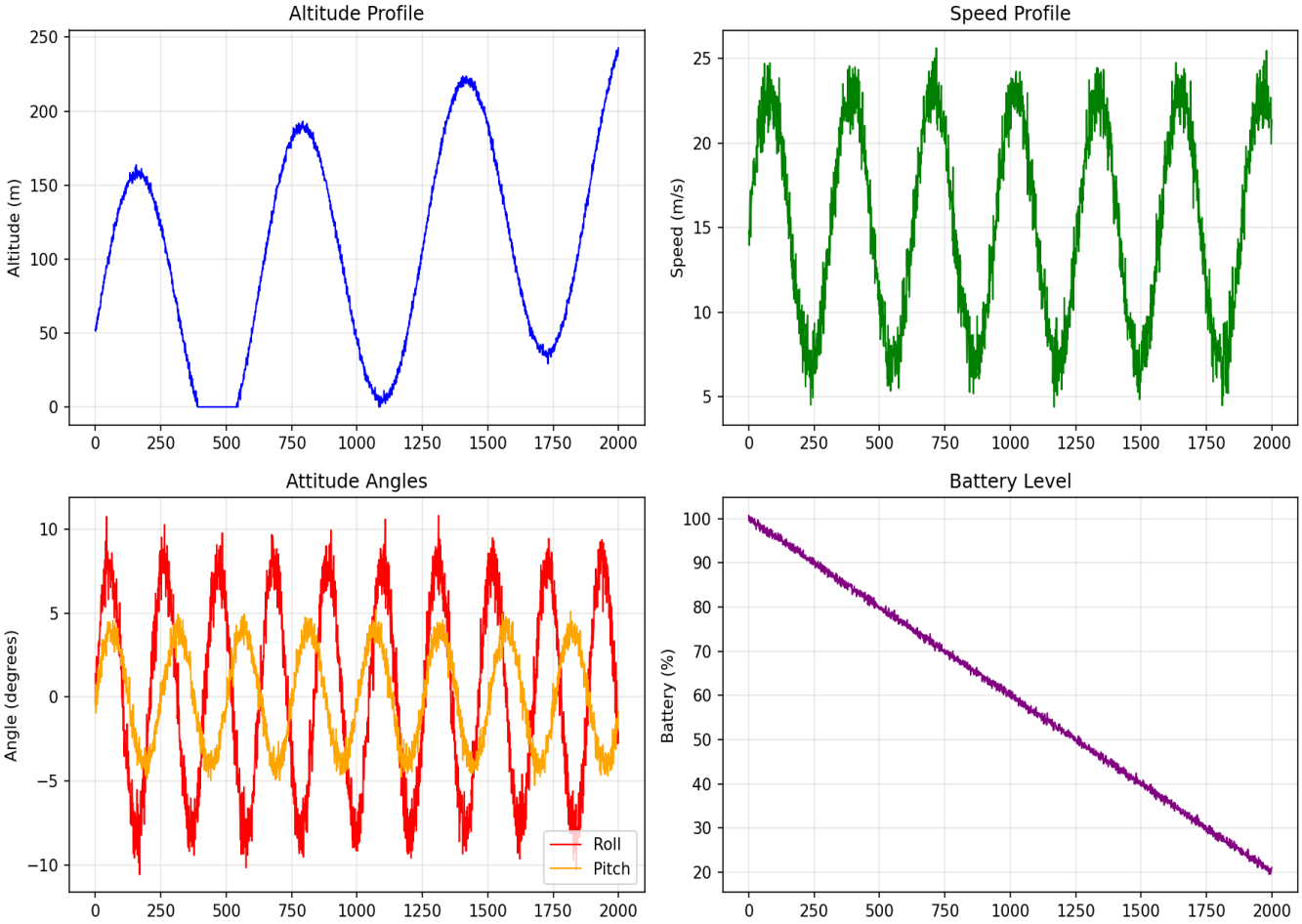
Flight Visualizations

The following visualizations provide comprehensive insights into the flight characteristics, including altitude profiles, speed variations, attitude changes, and battery consumption patterns.



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Conclusions and Recommendations

Flight Performance Conclusions

The flight demonstrated excellent mission execution with a duration of 33.3 minutes, covering 36376.0 meters at an average speed of 15.3 m/s. The aircraft successfully maintained controlled flight parameters throughout the operation, indicating reliable system performance and operational readiness. The combination of speed and altitude performance suggests optimal flight efficiency with effective power management and aerodynamic performance characteristics. High anomaly count (1302) with 65.1% anomaly rate indicates poor - high anomaly rate flight quality requiring immediate attention. Multiple systems show performance issues that could impact mission safety and effectiveness. Battery performance showed poor - very high consumption rate or many anomalies characteristics with a moderate consumption rate of 2.39% per minute. While acceptable for current operations, there are opportunities for power optimization to extend flight endurance and mission range. Flight stability assessment of excellent with a score of 1.0/100 suggests the need for control system optimization. The stability issues detected could impact mission precision and require immediate attention to ensure safe and reliable operations. Flight phase detection identified 6 distinct operational phases, demonstrating highly efficient mission execution. The clear phase segmentation indicates proper flight planning and systematic progression through mission objectives. The comprehensive flight analysis provides valuable insights into system performance, operational efficiency, and mission effectiveness. The data collected enables evidence-based decision making for maintenance scheduling, operational planning, and system optimization.

Recommendations

IMMEDIATE ACTIONS REQUIRED: • Schedule comprehensive system inspection and maintenance • Review and recalibrate all flight control systems • Implement enhanced pre-flight check procedures • Consider temporary operational restrictions until issues resolved **POWER EFFICIENCY IMPROVEMENTS:** • Review current flight patterns for efficiency opportunities • Optimize cruise speeds and altitude profiles • Consider aerodynamic improvements • Implement power-saving flight modes where appropriate **STABILITY IMPROVEMENTS:** • Conduct comprehensive flight control system calibration • Review and adjust PID controller parameters • Check for mechanical issues affecting stability • Implement enhanced stabilization systems • Consider pilot training for improved control inputs **CONTINUOUS IMPROVEMENT:** • Implement regular flight data analysis routines • Establish performance benchmarks and standards • Create comprehensive maintenance schedules • Develop pilot training programs based on analysis insights **SAFETY AND COMPLIANCE:** • Review and update safety procedures based on analysis findings • Ensure compliance with regulatory requirements • Implement risk mitigation strategies • Maintain detailed operational documentation **TECHNOLOGY UPGRADES:** • Consider advanced sensor integration for enhanced monitoring • Evaluate automated flight control systems • Implement real-time anomaly detection and alerting • Explore AI-powered flight optimization systems **DATA MANAGEMENT:** • Establish comprehensive data collection and storage systems • Implement automated analysis and reporting workflows • Create historical performance databases • Develop predictive maintenance algorithms

This comprehensive analysis provides valuable insights into the UAV flight performance and can be used for mission planning, pilot training, and system optimization. Regular analysis of flight data is recommended to maintain optimal performance and safety standards.